I got the power

Principled Bayesian dynamic borrowing with power prior methods

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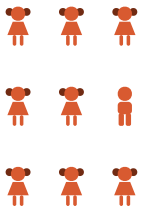
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Motivation

Paediatric RCT

PLUTO, n = 92



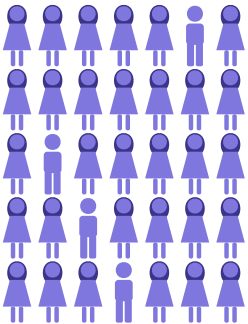
few, hard to recruit

borrow?

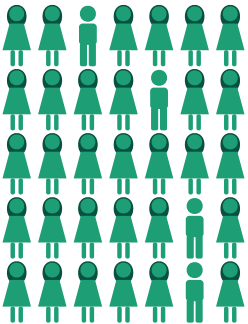
how much?

Adult trials

same drug, same endpoint



BLISS-52, n = 548



BLISS-76, n = 577

Setup

Let

- $\mathbf{D_0} = (D_0^{(1)}, \ldots, D_0^{(n_0)}) \sim P_0 \equiv P_{\theta_0}$ be the **historical** data;
- $\mathbf{D} = (D^{(1)}, \ldots D^{(n)}) \sim P_1 \equiv P_{\theta_1}$ be the **current** data.
- Model is well-specified

Power prior

$$\pi_{\text{PP}}^{(\eta)}(\theta|D_0) := \frac{L(D_0|\theta)^\eta \pi_0(\theta)}{c_0(\eta)},$$

Normalised power prior

$$\pi_{\text{NPP}}(\theta, \eta \mid D_0) := \pi_{\text{PP}}^{(\eta)}(\theta \mid D_0)\pi_A(\eta) \propto \frac{L(D_0 \mid \theta)^\eta}{c_0(\eta)} \pi_0(\theta)\pi_A(\eta).$$

Main objects

$$
\begin{aligned}
p_{\text{NPP}}(\theta, \eta \mid D_0, D) &:= L(D \mid \theta)\pi_{\text{NPP}}(\theta, \eta \mid D_0), \\
&\propto \frac{L(D \mid \theta)L(D_0 \mid \theta)^{\eta}}{c_0(\eta)}\pi_0(\theta)\pi_A(\eta), \\
&\propto p(\theta \mid D)L(D_0 \mid \theta)^{\eta}\frac{\pi_A(\eta)}{c_0(\eta)}.
\end{aligned}
$$

- $p^T_{\text{NPP}}(\theta \mid D_0, D) \equiv$ marginal effect; of central interest
- $p^A_{\text{NPP}}(\eta \mid D_0, D) \equiv$ tells us something about compatibility

Roadmap

"Coherent" borrowing

How do we make $\eta \to 0$ when $P_0 \neq P_1$ **and** $\eta \to 1$ when $P_0 = P_1$?

Principled computation

NPP is doubly-intractable; can we compute exactly? Are there **good**, efficient approximations? Can we sample **exactly**?

Multiple historical data sets

Strange things (can) happen when you have multiple historical data sets.

Oliveira et al. (2026+), Shen et al. (2026)[1]:

Theorem (NPP shrinks to zero under discrepancy)

If $P_0 \neq P_1$, under suitable conditions,

$$p^A_{\text{NPP}}(\eta \mid D_0, D) \to \delta_{\{0\}}(\eta),$$

a.s. $P^n_{\theta_1} \otimes P^{n_0}_{\theta_0}$.

Unwanted(?) discounting

However, when $P_0 = P_1$ the NPP does **not** concentrate at 1, and this is seen as a problem (Pawel et al. , 2023).

[1] and others.

Why though?

Theorem: inability to fully borrow

if π_A is continuous and positive on $(0, 1]$, then

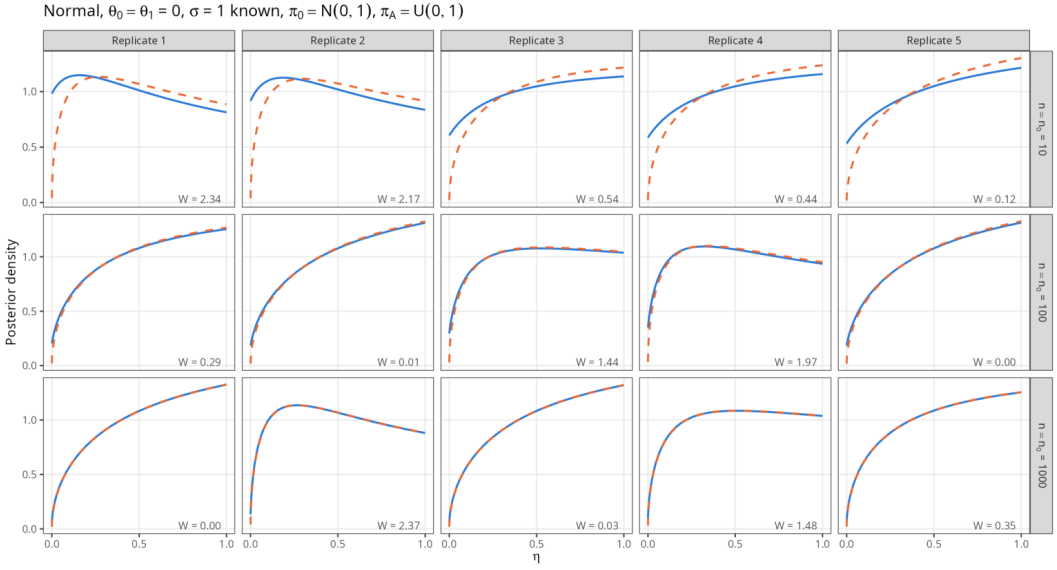
$$d_{\mathrm{TV}}\big(p_{\mathrm{NPP}}(\eta \mid D, D_0),\ g_n(\eta)\big) \xrightarrow{p} 0, \quad \text{whence}$$

$$\begin{aligned} g_n(\eta) &\propto \lambda_r(\eta)^{p/2} e^{-\lambda_r(\eta) W_n/2} \pi_A(\eta), \\ W_n &= (\hat{\theta}_1 - \hat{\theta}_0)^\top (J_1^{-1} + J_0^{-1})^{-1} (\hat{\theta}_1 - \hat{\theta}_0), \\ \lambda_r(\eta) &= \frac{\eta(1+r)}{1+\eta r}. \end{aligned}$$

In particular, the posterior does not converge to a point mass at $\eta = 1$.

Proof idea(s): under regularity assumptions, do a Laplace approximation of the marginal likelihood under compatibility.

Illustration: normal model with known variance



A couple of tentative solutions

Data-adaptive, fixed η

Choose η in a data-dependent manner, but keep it fixed

Information matching power prior (IMPP)

Matching information between the power prior and the regular posterior. This is an adaptation of Holmes & Walker (2017).

Predictive criterion power prior (PCPP)

Making the predictive criterion of Reznik (2026) work on real data.

Information matching power prior (IMPP)

We choose η such that

$$\int_{\mathfrak{X}^{k_n}} I_\eta(x) L(x \mid \theta_0) \mathrm{d}x = \int_{\mathfrak{X}^{k_n}} I_1(x) L(x \mid \theta_1) \mathrm{d}x,$$

where $k_n = \lfloor (rn)^\nu \rfloor = \lfloor n_0^\nu \rfloor$ for $\nu \in [0, 1]$, $I_\eta(x) = \eta^2 \Delta(x)$, and $\Delta(x)$ is the Fisher relative information divergence of the posterior update from $p(\theta \mid D)$ to $p(\theta \mid D, x)$.

$$\Delta(x) = \int \sum_{j=1}^{d} \left\{ \sum_{i=1}^{k_n} \frac{\partial}{\partial \theta_j} \log L(x_i \mid \theta) \right\}^2 p(\theta \mid D) \mathrm{d}\theta$$

This setup leads to (Braga & Oliveira et al. (2026+)):

Information matching power prior (IMPP)

$$\hat{\eta}_{\text{IM}} = \left(\frac{\int_{\mathcal{X}^{k_n}} \Delta(x) L(x \mid \hat{\theta}_1) \mathrm{d}x}{\int_{\mathcal{X}^{k_n}} \Delta(x) L(x \mid \hat{\theta}_0) \mathrm{d}x} \right)^{1/2}.$$

Fixed variance normal example

Suppose that $y_{0i} \sim \mathcal{N}(\theta_0, \sigma^2)$ with likelihood $L_0(y_{0i} \mid \theta_0, \sigma^2)$, and $y_i \sim \mathcal{N}(\theta_1, \sigma^2)$.

$$\eta = \left(\frac{\frac{v}{\sigma^2} + \frac{1}{\sigma^2}(m - \theta_1)^2 + \frac{1}{k_n}}{\frac{v}{\sigma^2} + \frac{1}{\sigma^2}(m - \theta_0)^2 + \frac{1}{k_n}} \right)^{1/2}.$$

Predictive criterion

Under a parametric model of dimension d, define

$$K_i(\theta, \vartheta) = D_{KL}(p_\theta(\cdot \mid x_{0i}), p_\vartheta(\cdot \mid x_{0i})),$$

historical design points.

Predictive criterion power prior (IMPP)

The theoretical optimum based on the full historical sample is

$$\eta^*_{\text{PC}} = \frac{d}{d + 2\sum_{i=1}^{n_0} K_i(\theta_0, \theta_1)}.$$

We consider using

$$\hat{\eta}_{\text{PC}} = \frac{d}{d + 2\sum_{i=1}^{k_n} K_i(\hat{\theta}_0, \hat{\theta}_1)}.$$

Here, have your "coherence"

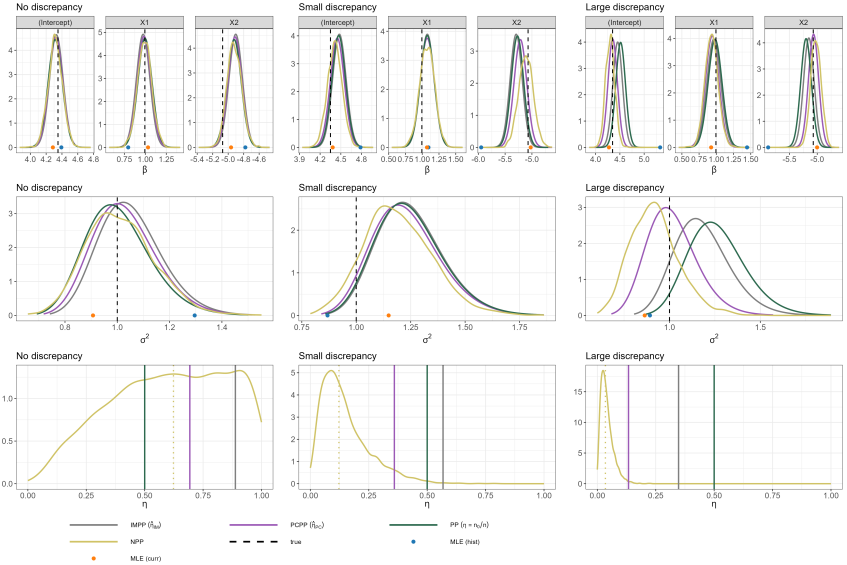
Theorem

Let $D \sim P_{\theta_1}^n$ and $D_0 \sim P_{\theta_0}^{n_0}$ with $r = n_0/n$ a fixed constant. Under appropriate conditions, for a given sequence $\{k_n\}_{n=1}^{\infty} \subset \mathbb{N}$ with $k_n \leq n_0$ we have the following implications:

(i) *If $k_n \to \infty$, then $\hat{\eta}_{\text{IM}} = O_p(k_n^{-1/2})$ whenever $\theta_0 \neq \theta_1$, and, in particular $\hat{\eta}_{\text{IM}} \xrightarrow{p} 0$;*

(ii) *If $k_n/n \to 0$, then $\hat{\eta}_{\text{IM}} = 1 + O_p(k_n/n + n^{-1/2})$ whenever $\theta_0 = \theta_1$, and, in particular, $\hat{\eta}_{\text{IM}} \xrightarrow{p} 1$.*

Results


No discrepancy
Small discrepancy
Large discrepancy
(Intercept)
X1
X2
β
σ²
η
IMPP ($\hat{\eta}_{IW}$)
NPP
MLE (curr)
PCPP ($\hat{\eta}_{DC}$)
true
PP ($\eta = n_0/n$)
MLE (hist)

Variational Normalised Power Prior (VNPP)

- Careful asymptotic analysis: we prove new and general BvM-type results for the PP and the NPP;
- This motivates a MVN variational approximation → VNPP

Theorem (Variational approximation under compatibility)

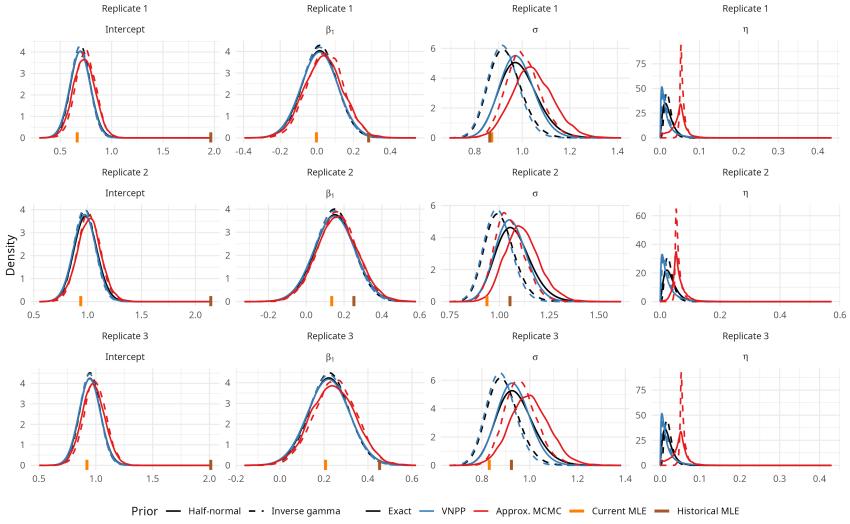
Suppose that $\theta_0 = \theta_1$. *Under appropriate conditions, if* $(1/n_0) \ll \varepsilon_{n_0} \ll 1$, *we have that*

$$\mathbb{P}\left[\int D_\eta(P_\theta, P_{\theta_0})\tilde{p}^{(\eta)}_{\mathrm{PP}}(\mathrm{d}\theta \mid D, D_0) \le c(r,\eta)\right] \ge 1 - \frac{1}{n_0\varepsilon_{n_0}}h(r), \text{ whence}$$

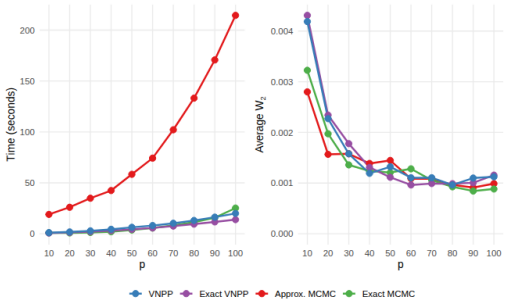
$$c(r,\eta) := 2\left(3 + \frac{1}{r}\right)\frac{1+\eta}{1-\eta}\varepsilon_{n_0},$$

$$h(r) := 1 + \frac{1}{r} + \frac{2}{3+1/r}.$$

VNPP works!


Replicate 1
Intercept
β1
σ
η
Replicate 2
Intercept
β1
σ
η
Replicate 3
Intercept
β1
σ
η
Density
Prior
Half-normal
Inverse gamma
Exact
VNPP
Approx. MCMC
Current MLE
Historical MLE

VNPP works! (2)



Parameter	Method	Bias	MSE	WIS	Coverage (95%)
β_0	Approx. MCMC	-0.001	0.005	0.034	0.966
	Exact MCMC	-0.001	0.005	0.034	0.968
	VNPP	-0.001	0.005	0.033	0.962
	Exact VNPP	-0.001	0.005	0.033	0.962
β_1	Approx. MCMC	-0.005	0.006	0.035	0.958
	Exact MCMC	-0.005	0.006	0.034	0.960
	VNPP	-0.005	0.006	0.034	0.960
	Exact VNPP	-0.005	0.006	0.034	0.960
σ	Approx. MCMC	-0.031	0.003	0.026	0.954
	Exact MCMC	-0.031	0.003	0.026	0.956
	VNPP	-0.027	0.003	0.025	0.938
	Exact VNPP	-0.027	0.003	0.025	0.940

Sampling exactly from the NPP: the exchange algorithm

The Exchange algorithm was first introduced by Murray, Ghahramani, and MacKay (2012) and extended for the NPP by Adame et al. (2026+).

- Given the current state (θ, η), propose $\eta' \sim q(\cdot \mid \eta)$.
- Draw an auxiliary parameter exactly from the proposed power prior:

$$\theta^\star \sim \pi_{\mathrm{PP}}^{(\eta')}(\cdot \mid D_0)$$

- The auxiliary draw allows the two intractable normalising constants to cancel in an augmented Metropolis–Hastings acceptance ratio.
- The resulting transition never evaluates $c_0(\eta)$ or $c_0(\eta')$.

Key distinction

θ is the current parameter value; $\theta^\star$ is an auxiliary draw used only in the acceptance decision.

Sampling exactly from the NPP: the exchange algorithm

The acceptance ratio for updating η, conditional on the current θ, is

$$R_{\text{ex}} = \frac{\pi_A(\eta')}{\pi_A(\eta)} \frac{q(\eta \mid \eta')}{q(\eta' \mid \eta)} \frac{L_0(D_0 \mid \theta)^{\eta'-\eta}}{\boxed{L_0(D_0 \mid \theta^\star)^{\eta'-\eta}}}.$$

- The intractable ratio $\boxed{c_0(\eta)/c_0(\eta')}$ disappears.
- The auxiliary likelihood term replaces the normalising-constant ratio.
- After the accept/reject decision, $\theta^\star$ is discarded.

The catch

The auxiliary draw must be exact from the power prior.

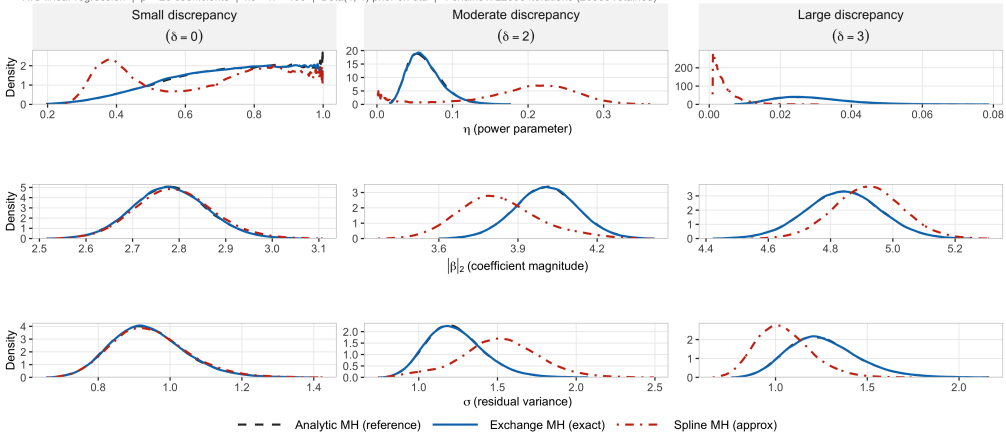
- Achievable in conjugate models, such as Normal–Inverse-Gamma regression.
- Not automatic for general non-conjugate models: ordinary finite-step MCMC draws are generally approximate.

The perks of avoiding the normalising constant

Exchange algorithm vs spline emulation for the normalised power prior

Analytical MH (reference) vs Exchange MH vs spline-emulated MH

NIG linear regression | p = 20 coefficients | n0 = n = 100 | Beta(1, 1) prior on eta | 4 chains x 22000 iterations (20000 retained)



	acc	ESS(eta)	Rhat	mean(eta): sm / mo / lg	W1(eta) vs ref: sm / mo / lg
Analytic MH (reference)	0.47	17884	1.000	0.728 / 0.064 / 0.029	reference
Exchange MH (exact)	0.46	5985	1.001	0.725 / 0.064 / 0.029	0.003 / 0.000 / 0.000
Spline MH (approx)	0.51	7146	1.001	0.656 / 0.192 / 0.004	0.072 / 0.130 / 0.025

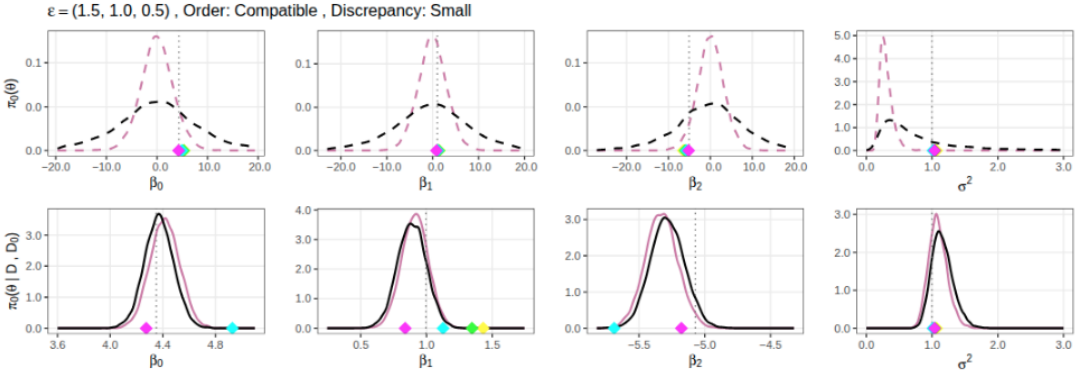
Normalisation matters

If you have data sets $D_{0k}, k = 1, 2, \ldots, K$, how do you actually do things? Jointly or sequentially?

$$\pi_{\text{NPP}}\left(\boldsymbol{\theta}, \boldsymbol{\eta} \mid \boldsymbol{D}_0\right) = \frac{\pi_A(\boldsymbol{\eta})\pi_0\left(\theta\right)\prod_{k=1}^{K} L_k\left(D_{0k} \mid \boldsymbol{\theta}\right)^{\eta_k}}{c_0\left(\boldsymbol{\eta}\right)},$$

$$\pi_{\text{NPP}_{\text{SEQ}}}\left(\boldsymbol{\theta}, \boldsymbol{\eta} \mid \boldsymbol{D}_0\right) = \pi_A(\boldsymbol{\eta}) \prod_{k=1}^{K} p_{\eta_k}\left(\boldsymbol{\theta} \mid D_{0k}\right).$$

Consequences!



Take home

It's possible to get "coherent" borrowing

But you give up on convergence speed

Variational approximations with guarantees

Beats the state-of-the-art in the accuracy-tractability trade-off

Multiple historical data sets

Are the future, but require careful prior elicitation.

Power priors provide semi-automatic learning

But other alternatives and connections (e.g. to hierarchical models) should be investigated – see Shen et al. (2026b) and Chen & Braga et al. (2026+)

THE END